

Calibration of Liquidity Parameters in the NWZ (2025) Framework

Covering equity market impact, options vol-space impact, and vol-point liquidity parameter estimation for a CRB / SPX Vega extension

Compiled from research discussions — June 2025

Context. Nutz, Webster & Zhao (2025) *"Unwinding Stochastic Order Flow"* models a Central Risk Book with three liquidity parameters: β (impact decay rate), λ (Kyle's lambda, price impact level), and ϵ (bid-ask spread cost). This review surveys the empirical and theoretical literature on estimating these parameters — first in standard equity units, then in **volatility-point units** required for a Vega-flow extension of the model.

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1. Lambda (λ): Price Impact Level — Equity Baseline

Kyle's lambda is the central measure of permanent-plus-transient price impact per unit of signed order flow. In the NWZ model it enters the cost functional and the opening block-trade formula (eq. 2.22), and drives the magnitude of the g_t feedback coefficient.

1.1 Canonical Definition and OLS Estimation

The standard estimator regresses price changes on signed volume:

$$\Delta P_t = \lambda \cdot Q_t + \text{noise}$$

A higher λ signals lower market depth. Estimation requires high-frequency trade-and-quote data; Kyle (1985) established the theoretical foundation linking λ to the adverse-selection component of the spread.

1.2 Large-Scale Empirical Benchmark: Almgren et al. (2005)

The most-cited empirical calibration study. Using approximately 700,000 US equity orders executed by Citigroup trading desks (December 2001 – June 2003), the authors fit market impact across a wide range of stocks as a function of volatility, average daily volume (ADV), and turnover. Key finding: temporary impact follows a 3/5 power law in trade rate (rejecting the square-root model for that component). Results are directly usable for pre- and post-trade cost estimation and optimal trade scheduling parameter inputs.

1.3 Machine-Learning Estimation from Daily Data

Dai et al. (2025, *Journal of Financial Markets*) combine existing microstructure models with raw stock-level features and machine learning to estimate Kyle's lambda (along with effective spread, realized spread, and mid-price impact) from daily OHLC data. Coverage: 992 US stocks over 5,000 trading days (2005–2024) and 2,081 Chinese stocks (2019–2024). This provides a practical approach for desks lacking TAQ-level data.

1.4 Implication for NWZ CRB

Lambda should be updated pre-market via rolling OLS of mid-price changes on signed vega flow (in vol-point units for the Vega extension). The effective $\lambda_{v,t}$ is a daily scalar that linearizes the proprietary impact function $\Lambda(\cdot)$ at the expected intraday operating range, preserving the LQ structure of the Riccati ODE system.

2. Beta (β): Transient Impact Decay Rate

Beta governs the rate at which the price impact state Y_t reverts to zero: $dY_t = -\beta Y_t dt + \lambda q_t dt$. In NWZ the default value is $\beta = 8 \text{ day}^{-1}$, corresponding to a half-life of approximately 40 minutes ($\ln 2 / 8$). This is the single most operationally important parameter: misspecification of β dominates calibration error, and under-estimation (slower-than-actual decay) is more costly than over-estimation.

2.1 Theoretical Foundation: Exponential Decay and No-Arbitrage

Gatheral et al. establish that exponential impact decay (the Obizhaeva-Wang kernel) is the unique decay shape compatible with linear market impact and the no-dynamic-arbitrage principle. This provides theoretical consistency for the NWZ exponential resilience model. Obizhaeva & Wang (2013) is the canonical model: each buy pushes price up proportionally and the dislocation reverts exponentially, producing block trades at open and close in the optimal strategy.

2.2 Empirical Reality: Power-Law vs. Exponential

Brokmann, Sérié, Kockelkoren & Bouchaud (2015) use a proprietary CFM metaorder dataset to deconvolve impact decay from autocorrelated order flow and return predictability over a 10-day horizon. Finding: impact decays to zero (or a small residual) following a **power-law**, not an exponential. The apparent plateau in raw data is fully explained by order-flow autocorrelation.

Bucci, Benzaquen, Lillo & Bouchaud (2019, ANcerno database) extend this with 8+ million institutional metaorders. At the end of execution day, impact averages $\sim 2/3$ of peak; subsequent decay follows a power-law intraday and an exponentially-truncated power-law multiday, converging to $\sim 1/2$ of same-day value at ~ 50 -day horizon. Impact decay occurs in *volume time*, not calendar time.

Practical consequence for NWZ: β is an *effective* exponential approximation to a fundamentally power-law process. The linearization is justified only within the intraday horizon ($T = 1 \text{ day}$) where the exponential and power-law curves are close. Daily recalibration of β_{eff} via NLS on post-execution mid-price paths is recommended.

2.3 LOB Resilience and Timescale Evidence

Empirical LOB studies show that bid-ask spread and best-level depth recover within 5–10 seconds after a market order, while limit-order intensity takes ~ 30 minutes. This is consistent with NWZ's default $\beta = 8 \text{ day}^{-1}$ (half-life $\sim 40 \text{ min}$): the *flow impact* relevant for CRB operation lives at the order-intensity timescale, not the spread timescale. Chen, Horst & Tran (2019) provide the most direct estimation framework, fitting transient impact parameters from public NASDAQ LOB data for 100 stocks.

2.4 Estimation Protocol for CRB

- Collect metaorder-level data: entry time, exit time, signed vega notional, and mid-implied-vol path post-execution.
- Fit: $\text{MidImpact}(\tau) = \lambda_0 \cdot \exp(-\beta\tau)$ via NLS for each underlying/bucket.
- Rolling window (recommend 20 trading days), stratified by tenor bucket (e.g., $<1\text{M}$, $1\text{--}3\text{M}$, $3\text{--}12\text{M}$) and moneyness.
- In equity space: $\beta \sim 5\text{--}15 \text{ day}^{-1}$ is typical. In vol space: expect slower decay ($\beta_v \sim 1\text{--}4 \text{ day}^{-1}$, half-life 4–17 hours) due to vol mean-reversion being slower than underlying LOB resilience.

3. Epsilon (ϵ): Bid-Ask Spread Parameter

The spread parameter ϵ enters the LQ cost as $\epsilon/2 \cdot q_t^2$, representing the instantaneous cost of trading at speed q . It is distinct from permanent or transient impact: it captures the round-trip spread cost incurred at each infinitesimal trade. In NWZ's default calibration, $\epsilon = 0.01$.

3.1 Classical Spread Estimators

Roll (1984): The foundational approach estimates the effective spread from the first-order autocovariance of price changes: $s = 2\sqrt{-\text{Cov}(\Delta P_t, \Delta P_{t-1})}$. Subsequent refinements — Hasbrouck (2009) Gibbs sampler, Holden (2009) EffTick, Corwin-Schultz (2012) — improve estimation from daily data when tick-level quotes are unavailable.

CHL Estimator (Abdi & Ranaldo, 2017, RFS): Uses daily close, high, and low prices. In cross-sectional correlation with TAQ effective spreads, CHL outperforms Roll, Gibbs, EffTick across the 1993–2021 CRSP-TAQ sample (monthly average correlation 0.74 vs. 0.38–0.65 for alternatives). Recommended when quote data is unavailable.

Holden & Jacobsen (2014): Document that MTAQ-based estimates produce distorted spreads; Daily TAQ (post-2004) is the preferred benchmark data source.

3.2 Derivative Hedge Theory for Options Spreads

George & Longstaff propose the *derivative hedge theory*: in a world of perfect hedging, option spreads arise from underlying market illiquidity rather than inventory risk or informed trading in the option market itself. Key finding: option volume is *not* a significant determinant of option spreads — underlying market duration matters more. Spreads are positively correlated with underlying bid-ask spreads, supporting the hedging-cost channel.

This implies: for vol-space ϵ_v , the relevant spread measure is the **implied volatility bid-ask spread** $(\text{ask_IV} - \text{bid_IV}) / 2$, measured in vol points. For SPX ATM options this is typically 0.1–0.5 vol points. The spread also scales with vega $\times$ underlying spread, consistent with the options microstructure literature.

3.3 SPX-Specific Spread Evidence

Cao, Jacobs & Ke ("Derivative Spreads: Evidence from SPX Options") document intraday SPX spread dynamics: a U-shaped pattern with peaks at open and close (consistent with market maker bargaining power), and significant order-imbalance effects. Spreads widen ahead of macro announcements, a feature not captured by time-average ϵ and motivating time-varying ϵ_t for the Vega model extension.

Eraker & Wu ("Market Maker Inventory, Bid-Ask Spreads") model SPX market-maker inventory dynamics and show that asking prices are more volatile than bids, driven by market maker net short gamma positioning — a structural asymmetry relevant for vol-space spread estimation.

3.4 Converting Spread to ϵ

In NWZ units [$\text{bps}^2 \cdot \text{day} / \text{ADV}\%^2$], the conversion from effective spread s (in bps) is approximately:

$$\epsilon \approx s / (2 \cdot q_{\text{typical}} \cdot T)$$

where q_{typical} is the typical intraday trading rate in ADV% and $T = 1$ day. For the vol-space extension, s is the IV bid-ask spread in vol points and q_{typical} is in vega ADV% units. Daily pre-market calibration using OPRA snapshot data is recommended.

4. Joint Calibration Frameworks

4.1 Webster (2023) Handbook

The *Handbook of Price Impact Modeling* (Kevin T. Webster, CRC Press, 2023) is the most comprehensive treatment of practical parameter estimation, written by one of the NWZ co-authors. It covers the full pipeline from raw data to optimal execution algorithm, including cross-sectional and time-series calibration of $(\beta, \lambda, \varepsilon)$ under the Obizhaeva-Wang framework. Chapters 3–5 are the primary reference for CRB implementation.

4.2 Webster & Westray (2024) TCA Framework

This paper shows how to use intraday price and trading data to improve trading cost estimates within the OW model, deriving analytical formulas for cost estimation with transient price impact. Key result: faster impact decay (β larger) leads to higher model noise (lower signal-to-noise ratio in impact estimation). General principle: reducing noise in TCA measurements applies across a wide range of transient impact models, including propagator models (Bouchaud et al., 2006).

4.3 Chen, Horst & Tran (2019) — LOB Calibration for 100 Stocks

Derives explicit solutions for portfolio liquidation with instantaneous, permanent, and transient impact combined (Graewe-Horst 2017 framework). Shows how to relate the different impact forms to limit order book microstructure and estimate parameters from public market data. Applies to high-frequency LOB data from 100 NASDAQ stocks, providing benchmarks for (β, λ) in equity markets.

5. Vol-Point Extension: Lambda in Implied Volatility Space

The Vega-flow extension of NWZ requires redefining all liquidity parameters in volatility-point units. Lambda becomes the price impact of vega flow on the implied volatility surface: $\Delta IV = \lambda_v \cdot Q_v$.

5.1 Tóth, Eisler & Bouchaud (2016) — Square-Root Law in Options

This is **the foundational empirical paper** for vol-space impact estimation. Using CFM's proprietary options metaorder dataset, the authors define the metaorder size as the **net vega notional** Q_v and the market total gross vega traded as V_v . Impact is measured in implied volatility space (vol points). The volatility-of-volatility σ_σ is defined as the standard deviation of daily IV returns, weighted by traded vega. Key finding: the square-root impact law holds in options markets:

$$I(\tau) \sim Y_{\text{vol}} \cdot \sigma_\sigma \cdot \sqrt{(Q_v / V_v)}$$

The universal constant Y_{vol} is quantitatively similar to that observed across equity, futures, and FX markets, supporting a universal microstructure mechanism. This paper establishes that **vega notional is the correct unit for options order flow**, analogous to share volume in equity.

Implication for CRB: $\lambda_{v,t}$ should be estimated from OLS of delta-IV on signed vega flow, aggregated across all options on the same underlying (not contract-by-contract). Pre-market linearization of the proprietary $\lambda_{v,t}$ at the expected operating range gives an effective scalar $\lambda_{v,t}$, preserving LQ structure.

5.2 Said et al. (2019) — Systematic Study of HF Options Impact

Extends the Toth et al. result with finer resolution. Using a proprietary metaorder database, the analysis separates options flow into *ATM vol metaorders* (buying/selling ATM straddles — move level) and *ATM skew metaorders* (buying calls / selling puts at same strike — move skew). Square-root law holds for both with exponents ~ 0.56 and ~ 0.53 respectively. Impact is projected onto SVI vol surface parameters, revealing that vol flow impacts **both level and skew simultaneously**.

This implies λ_v is not isotropic on the vol surface. A CRB with significant skew flow exposure should distinguish:

- $\lambda_{v,\text{level}}$: impact of net vega on ATM vol
- $\lambda_{v,\text{skew}}$: impact of net risk-reversal vega on skew

For a first-order scalar model (current NWZ extension), λ_v is an ADV-weighted average. The directional decomposition is a natural next step toward full vol surface impact modeling.

6. Vol-Point Extension: Cross-Impact and Surface Deformation

6.1 Tomas, Mastromatteo & Benzaquen (2021/2023) — Cross-Impact in Derivatives

The most theoretically rigorous treatment of impact in derivative markets. The framework considers a universe of derivatives whose prices are deterministic functions of stochastic factors (underlying spot, implied vol level). Trades on both factors and derivatives induce impact, and the key insight is:

- In an efficient market, derivatives cannot respond to order flow *independently* of their underlying factors — any apparent instrument-level impact must be traced back to a small number of factors.
- The fragmented liquidity of many individually illiquid options must be *aggregated into liquidity pools* along the underlying factors to obtain a well-posed model.
- A particular cross-impact model — the multivariate Kyle model — prevents arbitrage and provides tractable estimation.

Empirical validation uses E-Mini futures, European call and put options, and VIX futures. The framework is applied to estimate cross-impact and shows it captures part of option return variation. Relevant for CRB as a principled justification for the **vega aggregation approach**: mapping heterogeneous option flows to a scalar vol factor flow.

6.2 Said (2019) — Hedging Metaorders and Impact Transmission

Studies the execution process of hedging derivatives via perturbation theory. Shows how delta-hedging metaorders shape market impact: the arbitrage opportunities present in discrete time vanish as trading frequency increases, yielding a pricing PDE. Relevant because for a CRB, much of the options flow is eventually delta-hedged in the underlying — the feedback from delta-hedging into vol impact is the mechanism by which β_v operates at a different timescale than equity β .

6.3 Multivariate Kyle and Liquidity Aggregation

Garcia del Molino, Mastromatteo, Benzaquen & Bouchaud (2018/2019) establish the micro-foundations of the multivariate Kyle model used by Tomas et al. Key result: aggregating option flows into vol factor flows is not just a practical simplification — it is the *theoretically correct* representation when derivatives are priced by no-arbitrage off the underlying. The liquidity of the vol surface is essentially the liquidity of the vol factor, not the sum of individual option liquidities.

7. Vol-Point Extension: Beta and Epsilon in Vol Space

7.1 Beta in Vol Space (β_v) — Literature Gap

There is **no published empirical study** directly estimating the transient vol-space impact decay rate β_v . This is a significant literature gap and represents an original contribution of the Vega-flow NWZ extension.

The decay of vol impact is mechanistically different from equity impact decay:

- **Delta-hedging feedback:** Options market makers delta-hedge continuously. A vol impact (IV shock) causes a vanna/charm adjustment that feeds back into underlying price, which in turn affects IV. This creates a feedback loop not present in equity.
- **Vol mean-reversion:** Implied vol is a stationary process with its own mean-reversion speed (theta in OU model terms). The vol surface mean-reverts at a timescale of days-to-weeks, much slower than equity LOB resilience (minutes).
- **Term structure effects:** Vol impact decays faster for short-dated options (high gamma/theta) than long-dated ones. β_v is likely tenor-dependent.

Practical recommendation: Estimate β_v from desk execution data: for each vega metaorder, collect the ATM IV mid path for 60-120 minutes post-completion and fit $IV_impact(\tau) = \lambda_0 \cdot \exp(-\beta_v \cdot \tau)$ via NLS. Stratify by tenor bucket. Expected range: $\beta_v \sim 1-4 \text{ day}^{-1}$ (half-life 4-17 hours). This is the primary empirical contribution of the Vega extension.

7.2 Epsilon in Vol Space (ϵ_v) — SPX Spread Evidence

The vol-space spread parameter ϵ_v corresponds to the instantaneous cost of trading vega at speed q_v , in units of vol points. The relevant spread measure is the **IV bid-ask spread**.

SPX-specific evidence (Cao, Jacobs & Ke): ATM SPX options have normalized quoted spreads with strong intraday pattern — U-shaped with peaks at open and close, consistent with market maker risk management. Dollar spreads vary substantially across moneyness and expiry; normalized (IV) spreads are more homogeneous. Order imbalances drive spreads, especially in the last 2 hours of trading.

George & Longstaff's derivative hedge theory implies: IV spread is approximately proportional to delta-hedging rebalancing cost = vega $\times$ underlying IV spread. For SPX ATM options, the implied vol bid-ask spread is typically **0.1 – 0.5 vol points** (10–50 bps in vol units), varying by expiry and moneyness.

Conversion to ϵ_v :

$$\epsilon_v \approx s_{IV} / (2 \cdot q_{v, \text{typical}} \cdot T)$$

where s_{IV} is the ATM IV spread in vol points and $q_{v, \text{typical}}$ is the desk's typical vega trading rate in vega-ADV% units.

8. Summary Table and Calibration Gaps

Parameter	Key References	Estimation Method	Coverage Quality
λ (equity)	Kyle (1985); Almgren et al. (2005); Dai et al. (2025)	OLS: $\Delta P \sim Q_{\text{signed}}$; rolling window; cross-sectional regression on ADV, vol	★★★★ Strong
β (equity)	Obizhaeva-Wang (2013); Brokmann et al. (2015); Bucci et al. (2019); Chen et al. (2019)	NLS fit to post-metaorder mid-price decay path; LOB resilience time	★★★ Good (power-law vs. exp tension)
ε (equity)	Roll (1984); Abdi-Rinaldo 2017 (CHL); Dai et al. (2025)	Roll autocovariance; CHL OHLC estimator; TAQ effective spread	★★★★ Strong
λ_v (vol space)	Tóth et al. (2016); Said et al. (2019); Tomas et al. (2021)	OLS: $\Delta IV \sim \text{signed vega notional}$ (aggregated); pre-market linearization of $\Lambda(\cdot)$	★★★ Good (CFM data, no public CRB study)
β_v (vol space)	Literature gap — no direct study	NLS on desk post-execution IV path; stratify by tenor/moneyness	★ Gap (original contribution)
ε_v (vol space)	Cao et al. (SPX); George-Longstaff; Eraker-Wu	IV bid-ask spread from OPRA; derivative hedge theory for spread decomposition	★★★ Partial (no standardized protocol)

Key Takeaways

- **Equity λ and ε are well-covered** by standard microstructure literature; direct translation to CRB requires desk-level regression.
- **Beta is the hardest to calibrate:** true decay is power-law; exponential approximation requires daily NLS re-estimation. Under-estimating β (slower-than-actual decay) is systematically more costly than over-estimation.
- **Vega-space λ_v** is established by Tóth et al. (2016) using vega-notional aggregation; the pre-market OLS linearization of the desk's proprietary $\Lambda_v(\cdot)$ is the correct implementation.
- **Beta_v is the primary literature gap** — no published empirical estimate exists for SPX options. This is an original contribution of the Vega-NWZ extension and should be estimated from desk metaorder data.
- **Epsilon_v** can be approximated from OPRA IV quotes; vol-space spread is more homogeneous when normalized by ATM IV than in dollar terms.
- The Tomas-Mastromatteo-Benzaquen (2021) cross-impact framework provides the **theoretical justification for aggregating heterogeneous option flows** into a scalar vol factor flow — the correct underpinning for the scalar λ_v approach.

9. References

Equity Market Impact and Lambda

- [Kyle 1985] Kyle, A.S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335.
- [Almgren et al. 2005] Almgren, R., Thum, C., Hauptmann, E. & Li, H. (2005). Direct estimation of equity market impact. *Risk*, 18, 57–62.
- [Dai et al. 2025] Dai, Y. et al. (2025). Estimating market liquidity from daily data: Marrying microstructure models and machine learning. *Journal of Financial Markets*, doi:10.1016/j.finmar.2025.101019.
- [Roll 1984] Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance*, 39(4), 1127–1139.
- [Abdi & Rinaldo 2017] Abdi, F. & Rinaldo, A. (2017). A simple estimation of bid-ask spreads from daily close, high, and low prices. *Review of Financial Studies*, 30(12), 4437–4480.

Beta / Impact Decay

- [Obizhaeva & Wang 2013] Obizhaeva, A. & Wang, J. (2013). Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets*, 16(1), 1–31.
- [Gatheral et al.] Gatheral, J. et al. Exponential resilience and decay of market impact. Demonstrates that exponential decay is compatible with linear impact and no-dynamic-arbitrage. *Quantitative Finance*.
- [Brokmann et al. 2015] Brokmann, X., Sérié, E., Kockelkoren, J. & Bouchaud, J.-P. (2015). Slow decay of impact in equity markets. *Market Microstructure and Liquidity*, 1(02), 1550007.
- [Bucci et al. 2019] Bucci, F., Benzaquen, M., Lillo, F. & Bouchaud, J.-P. (2019). Slow decay of impact in equity markets: Insights from the ANcerno database. *Market Microstructure and Liquidity*, 4(03n04), 1950006.
- [Chen et al. 2019] Chen, Y., Horst, U. & Tran, H.H. (2019). Portfolio liquidation under transient price impact: theoretical solution and implementation with 100 NASDAQ stocks. arXiv:1912.06426.

Epsilon / Bid-Ask Spread

- [Holden & Jacobsen 2014] Holden, C.W. & Jacobsen, S. (2014). Liquidity measurement problems in fast, competitive markets: expensive and cheap solutions. *Journal of Finance*, 69(4), 1747–1785.
- [Cao, Jacobs & Ke] Cao, J., Jacobs, K. & Ke, S. Derivative Spreads: Evidence from SPX Options. AFA Conference 2024 Working Paper.
- [George & Longstaff] George, T.J. & Longstaff, F.A. Option bid-ask spread and liquidity. NBER Working Paper 7331.
- [Eraker & Wu] Eraker, B. & Wu, Y. Market maker inventory, bid-ask spreads, and the computation of option-implied measures. Working paper.

Joint Calibration

- [Webster 2023] Webster, K.T. (2023). *Handbook of Price Impact Modeling*. CRC Press / Chapman & Hall. ISBN 978-1032328225.
- [Webster & Westray 2024] Webster, K.T. & Westray, N. (2024). Optimizing broker performance evaluation through intraday modeling of execution cost. arXiv:2405.18936.
- [NWZ 2025] Nutz, M., Webster, K. & Zhao, C. (2025). Unwinding stochastic order flow: When to warehouse trades. Columbia University Working Paper. GitHub: KevinThomasWebster/UnwindingStochasticOrderFlow.

Vol-Space / Options Market Impact

- [Tóth et al. 2016] Tóth, B., Eisler, Z. & Bouchaud, J.-P. (2016). The square-root impact law also holds for option markets. *Wilmott*, 2016. arXiv:1602.03043.
- [Said et al. 2019] Said, E. et al. (2019). Market impact: A systematic study of the high frequency options market. arXiv:1902.05418. [ATM vol and skew metaorder decomposition.]
- [Said 2019b] Said, E. (2019). How option hedging shapes market impact. arXiv:1910.05056. [Delta-hedging metaorder impact via perturbation theory.]

- [**Tomas et al. 2021**] Tomas, M., Mastromatteo, I. & Benzaquen, M. (2021/2023). Cross impact in derivative markets. arXiv:2102.02834; *Risk*, 2023. [Multivariate Kyle model; liquidity aggregation for options.]
- [**Garcia del Molino et al. 2018**] Garcia del Molino, L.C., Mastromatteo, I., Benzaquen, M. & Bouchaud, J.-P. (2018). The multivariate Kyle model: More is different. arXiv:1806.07791.

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